SimCRC Chile and USA calibration update

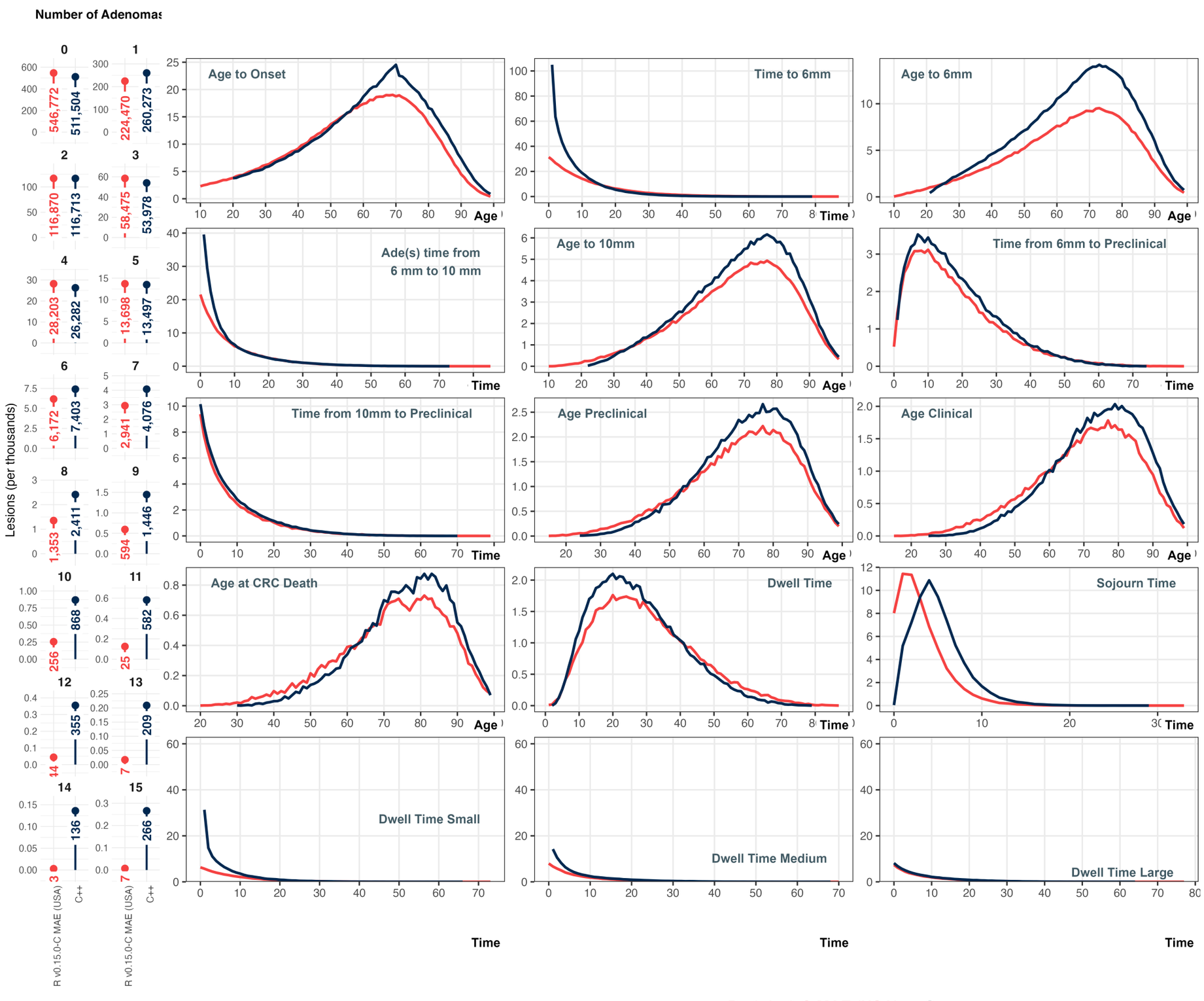
Updated USA calibration: 26 October 2026

SimCRC team

2026-11-05

Validation of C++ and SimCRC R Natural History Versions

Population: 1 million people
Generated on: 05-Nov-2026



Mean

Age to Lesion Onset
59.2 | 62.5

Time to 6mm
12.0 | 7.8

Age to 6mm
64.4 | 65.8

Ade(s) time from 6mm to 10mm
9.9 | 8.7

Age to 10mm
68.0 | 69.0

Time from 6mm to Preclinical
18.3 | 18.0

Time from 10mm to Preclinical
9.6 | 9.1

Age to Preclinical CRC
70.9 | 72.5

Age to Clinical CRC
71.1 | 74.0

Age to CRC Death
73.1 | 76.1

Dwell Time
30.3 | 27.4

Sojourn Time
3.6 | 5.0

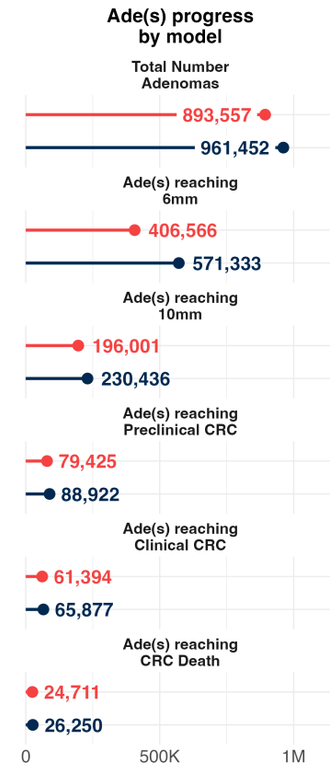
Phases dwell time

Small adenoma (<6 mm)
9.32 | 5.39

Medium adenoma (6-10 mm)
8.25 | 8.40

Large adenoma before cancer
9.17 | 8.57

Cancer before diagnosis (sojourn)
3.59 | 5.00



R v0.15.0-C MAE (USA) C++

Phase cases: R 61,394 | C++ 65,877

Calibration inputs and scope

Parameter source	Previous comparison	Updated comparison
USA	Published set v0.13.0.1	26 October 2026, v0.15.0
Chile	21 September 2026, v0.15.0	Same Chile set

The cheat sheet compares the updated USA set with the September Chile set.

Natural history: one million simulated women per country, without screening.

Life tables: Chile 2017 and USA 1977, preserving the original cheat-sheet assumptions.

Both countries use the female adenoma model. Adenomas are precancerous polyps. The cheat sheet uses the parameter sets selected for the smallest absolute calibration error.

Chile calibration

21 September 2026
v0.15.0

Numerical targets

Incidence per 100,000.

Prevalence as proportions.

Four parallel calibration sampling runs.

Grey: targets and fitting bounds.

Blue: calibrated model simulations and their spread.

Chile calibration

21 September 2026
v0.15.0

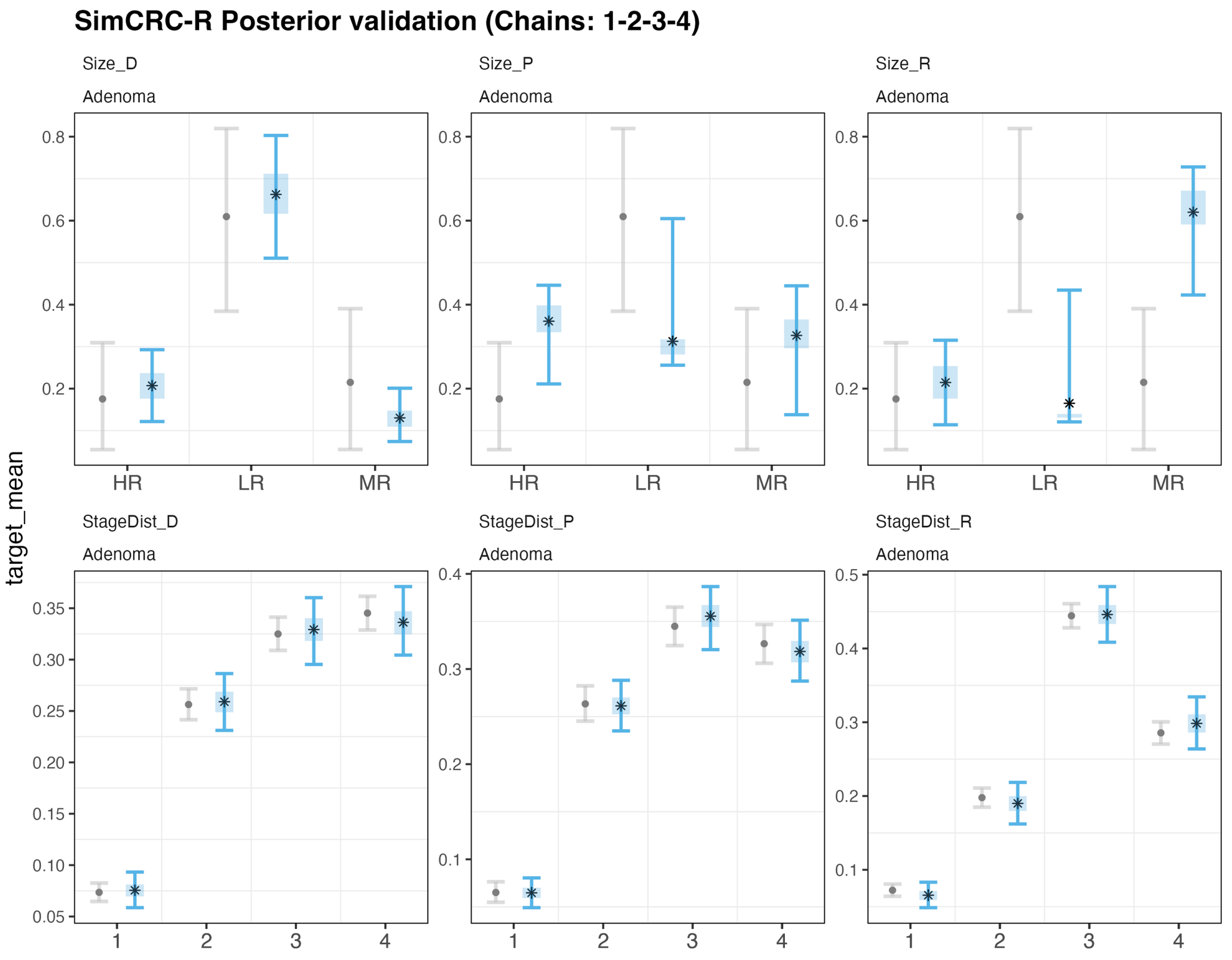
Categorical targets

Polyp size and cancer-stage proportions.

Four parallel calibration sampling runs.

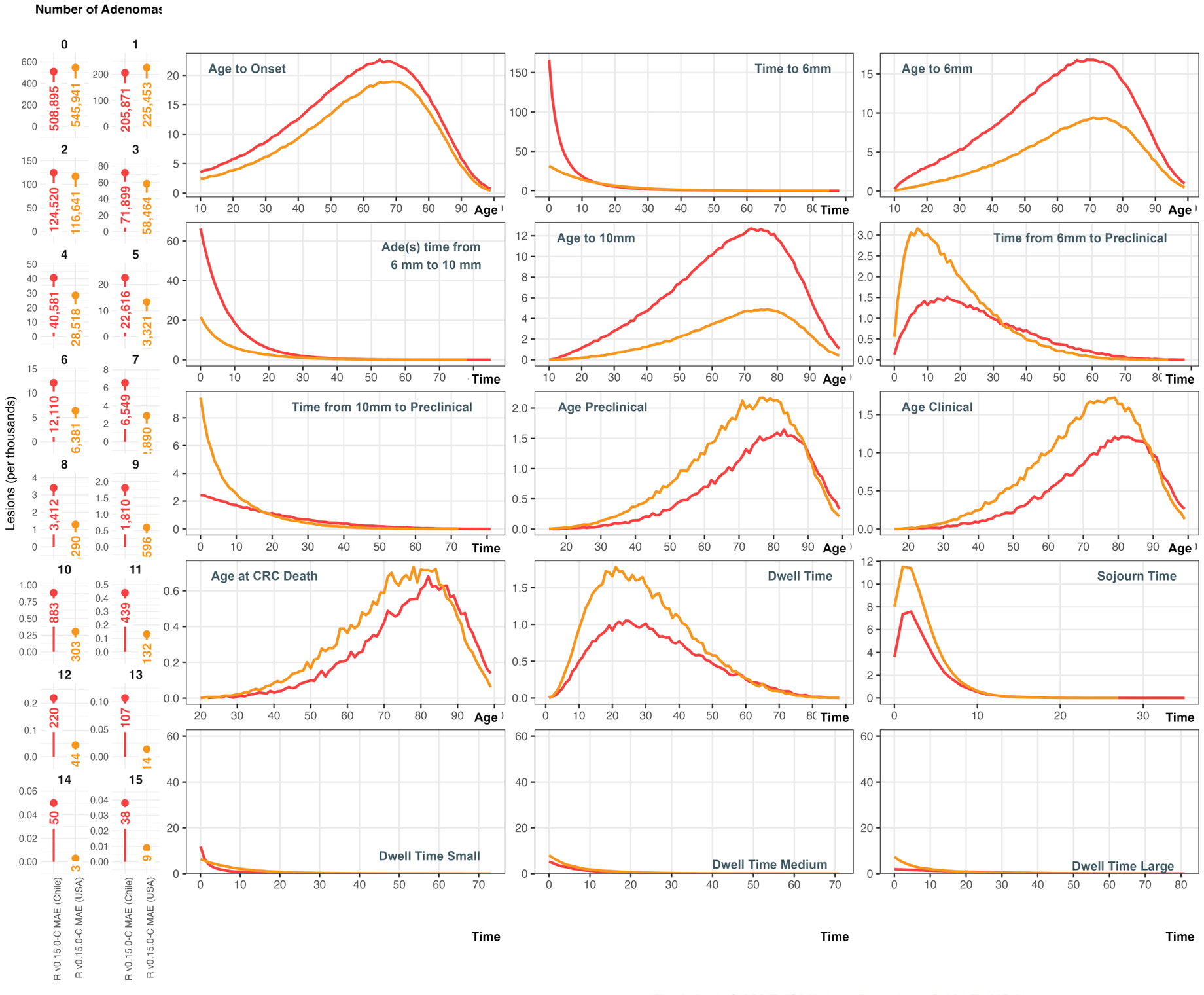
Grey: targets and fitting bounds.

Blue: calibrated model simulations and their spread.



SimCRC Chile and USA v0.15.0

Population: 1 million people
Generated on: 05-Nov-2026



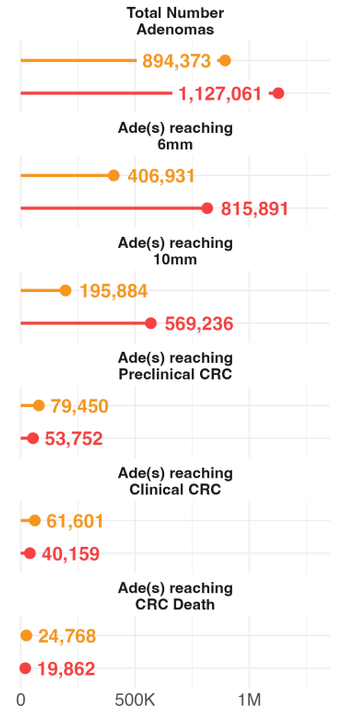
Mean

	R v0.15.0-C MAE (Chile)	R v0.15.0-C MAE (USA)
Age to Lesion Onset	58.0	59.2
Time to 6mm	6.6	12.0
Age to 6mm	61.4	64.4
Ade(s) time from 6mm to 10mm	8.4	10.0
Age to 10mm	65.1	68.0
Time from 6mm to Preclinical	25.5	18.3
Time from 10mm to Preclinical	17.3	9.6
Age to Preclinical CRC	75.2	70.9
Age to Clinical CRC	75.6	71.1
Age to CRC Death	77.0	73.0
Dwell Time	33.1	30.4
Sojourn Time	4.0	3.6

Phases dwell time

	R v0.15.0-C MAE (Chile)	R v0.15.0-C MAE (USA)
Small adenoma (<6 mm)	5.03	9.36
Medium adenoma (6-10 mm)	7.82	8.26
Large adenoma before cancer	16.32	9.16
Cancer before diagnosis (sojourn)	3.96	3.58

Ade(s) progress by model

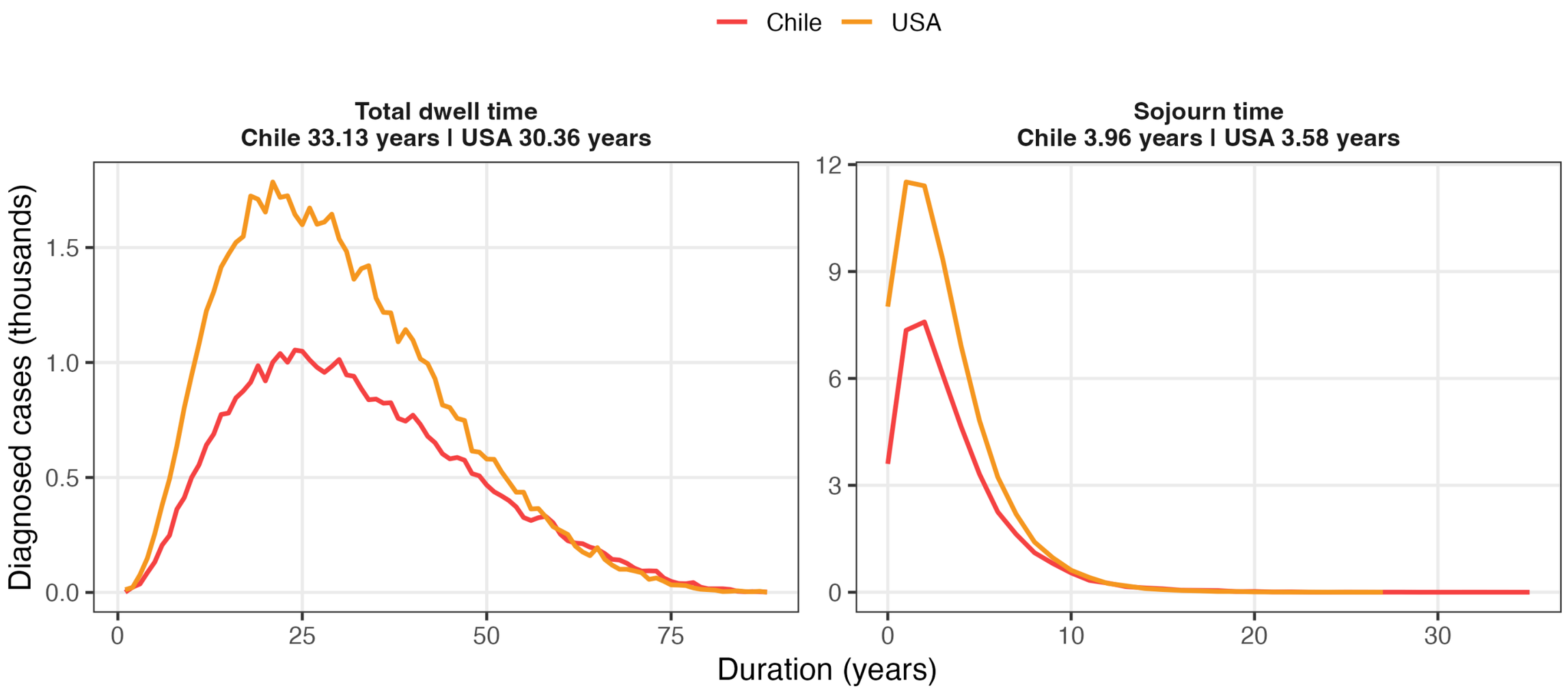


R v0.15.0-C MAE (Chile) R v0.15.0-C MAE (USA)

Phase cases: Chile 40,159 I USA 61,601

Dwell time and sojourn time

Total dwell: polyp formation to symptomatic diagnosis. **Sojourn:** cancer onset to symptomatic diagnosis.



Means describe the polyp that triggered diagnosis: 40,159 Chile cases and 61,601 USA cases.

Phases of dwell time

Each interval is calculated within the same diagnosing polyp before averaging across cases.

Phase	Interval	Chile, years	USA, years
Small adenoma	Polyp onset to 6 mm	5.03	9.36
Medium adenoma	6 mm to 10 mm	7.82	8.26
Large adenoma	10 mm to cancer onset	16.32	9.16
Sojourn	Cancer onset to diagnosis	3.96	3.58
Total dwell	Polyp onset to diagnosis	33.13	30.36

Chile’s longer total interval mainly comes from the **large-adenoma phase**, partly offset by its shorter early growth phases.

The four phases sum to total dwell time. The general growth panels include different groups of polyps.